Hazidien Ramadhan Utomo

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Summary

Urban and Regional Planning graduate from Institut Teknologi Sepuluh Nopember (ITS) with experience in spatial analysis, GIS, and remote sensing. Applied ArcGIS, ArcPy, and Global Mapper to assess rooftop solar potential across approximately 786,527 buildings in 31 districts of Surabaya, integrating LiDAR-derived data and solar radiation modelling. Experienced in land-use change modelling and transportation planning, with additional strengths in analytical problem-solving, team leadership, and coordination.

Education

Institut Teknologi Sepuluh Nopember — **Surabaya, Indonesia**
Fresh Graduate in Urban and Regional Planning | GPA: *3.36/4.00* — Aug 2022 – Aug 2026
Relevant Focus: GIS & Spatial Analysis, Remote Sensing, Site Planning, Regional and Urban Planning (RTRW/RDTR), Transportation Planning, and AutoCAD Based Road Corridor Planning.
Achievement:

- 3rd Place in the National GIS Thematic Infographic Research Competition (2024)

Work Experiences

Laboratory of Transportation and Spatial Analysis *(Contract)* — **Aug 2025 - Dec 2025**
Assistant Lecturer

- Delivered laboratory sessions for 41 students in the class. Guiding students through quantitative, qualitative, and GIS-based planning analysis, incorporating AI assisted learning tools to support clear and structured explanations.
- Presented and explained technical concepts, including spatial regression, clustering, and spatial autocorrelation.
- Facilitated discussions, responded to students' questions, and provided feedback to support discussion based learning.

PT Greenova Daya Prima *(Internship)* — **Jul 2025 - Aug 2025**
Renewable Energy Spatial Analysis Assistant

- Assessed rooftop solar potential using GIS based solar radiation analysis and LiDAR derived building data to support urban renewable energy planning.
- Integrated building, electricity consumption, land-use, and energy indicator datasets for the Surabaya Building Energy Baseline Study.
- Conducted field surveys covering 42 residential buildings and 12 commercial buildings and contributed to survey data validation and spatial modelling.

MRT Surabaya Feasibility Study — **Jun 2024 - Nov 2024**
Surveyor

- Conducted household, employee, and administrative surveys to collect travel and mobility data for transportation planning.
- Performed traffic counts and vehicle speed surveys, and analysed traffic performance using the applicable Indonesian road capacity assessment guidelines (MKJI/PKJI).
- Compiled survey data and carried out the four-step transportation modelling process, covering trip generation, trip distribution, modal split, and traffic assignment.

Selected Projects

Rooftop Solar PV Potential Assessment in Surabaya — **Sep 2025 - Jun 2026**

- Integrated building footprint shapefiles with LiDAR-derived elevation rasters processed in Global Mapper using ArcPy, then modelled rooftop solar radiation with ArcGIS Area Solar Radiation across 786,527 buildings in 31 Surabaya districts.
- Validated modelled solar radiation estimates against SolarGIS data, obtaining an RMSE of 143.40 kWh/m²/year and a reported relative error of 7.36%; estimated average annual PV electricity production of 291.32 kWh per m² of panel area.

Spatio Temporal Deforestation Modeling Project **Nov 2025 – Dec 2025**

- Developed spatio-temporal deforestation analysis workflows using Google Earth Engine (GEE) and multi-temporal Landsat imagery from 2009-2024 for environmental assessment.
- Evaluated and refined AI-assisted GEE workflows by reviewing analytical logic, parameter choices, and spatial results to improve the reliability of deforestation analysis.

Land Use Change Modelling **Jan 2025 - Feb 2025**

- Conducted land use change modelling using GeoSOS-FLUS based on 2018 and 2023 land use data and developed scenarios for 2023-2043 using ANN and Markov Chain analysis.
- Evaluated model performance using Kappa statistics and Figure of Merit (FoM) to assess spatial agreement and the results of land-use change simulations.

Agropolitan Regional Planning and Development Project **Aug 2024 – Dec 2024**

- Conducted regional assessment of agricultural potential, infrastructure needs, leading commodities, and service centres, supported by cadastral mapping and spatial analysis.
- Engaged in stakeholder discussions and Focus Group Discussions (FGDs) to identify local conditions, development needs, and planning priorities.

Leadership & Organizational Experiences

UKM Sepakbola ITS **Jan 2024 - Dec 2024**

Head of the Football Division

- Led and coordinated football training activities for an average of 42 active members, supporting team development and training participation.
- Collaborated with the coach in player selection and helped prepare an 18-player first-team squad for regional competition.
- Contributed to team preparation and achieved 1st Place in the Surabaya City Regional Football Trofeo UINSA competition.

Aksi Sepuluh Nopember 2023 **Aug 2023 - Dec 2023**

Vice Chairman I – Campus Orientation Event

- Coordinated four committees covering equipment, publication and documentation, events, and security and permits, overseeing approximately 40-50 committee members.
- Supervised cross functional coordination, event logistics, and operational readiness throughout event preparation and execution.

Certifications

- **Anthropic:** Claude 101 – AI Collaboration & Responsible AI Use (2026)
- **Aksara Lab Indonesia:** Intensive Bootcamp – LiDAR Data Classification & Contour Generation (2025)
- **Geoaccess Indonesia:**
  1. Thematic GIS Training on Land Cover Change Prediction Using QGIS and GeoSOS-FLUS (2025)
  2. Thematic GIS Training on Flood Modeling Based on Land Use Change and Prediction (2024)
  3. Basic and Thematic Google Earth Engine Training on Deforestation Impact on Air Pollution Analysis (2024)

Core Skills

Hard Skill: GIS, AutoCAD, Google Earth Engine, GeoSOS-FLUS, Global Mapper, HEC-RAS, Python, Canva, Word
Soft Skill: Communication, Teamwork, Problem Solving, Fast learning, Leadership, Problem Identification, Adaptability
Languages: Indonesian (Native), English (Professional Working Proficiency), Javanese (Native)
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